

Q1(a): $\begin{bmatrix} 3 & -6 & 9 \\ -2 & 7 & -2 \\ 0 & 1 & 5 \end{bmatrix}$

Row Echelon form

03 marks

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 5 \\ 0 & 0 & 33 \end{bmatrix}$$

Expansion by Row/column 01 mark

33 = value of determinant 01 mark

Q1(b): $8x - 4z + 5 = 2y$; $x = \frac{z}{2} + \frac{y}{4}$

$P_1: 8x - 2y - 4z + 5 = 0$; $P_2: 4x - y - 2z = 0$ 01 mark

$n_1 = (8, -2, -4)$; $n_2 = (4, -1, -2)$ 01 mark

$n_1 = 2n_2 \Rightarrow$ planes are parallel 01 mark

A point on plane P_2 ; put $x=0=y$

$\Rightarrow z=0$

point $(0,0,0)$

01 mark

$$\text{Distance} = \frac{|8(0) - 2(0) - 4(0) + 5|}{\sqrt{8^2 + (-2)^2 + (-4)^2}} = \frac{5}{\sqrt{84}} = \frac{5}{2\sqrt{21}} \text{ units}$$

01 mark

$$\text{Q2(a): } \{ \underset{v_1}{(2, -3, 1)}, \underset{v_2}{(4, 1, 1)}, \underset{v_3}{(0, -7, 1)} \}$$

$$\text{let } b = (b_1, b_2, b_3) \in \mathbb{R}^3 \neq 0$$

$$c_1(2, -3, 1) + c_2(4, 1, 1) + c_3(0, -7, 1) = (b_1, b_2, b_3)$$

01 mark

$$\det(A) = 0$$

01 mark

Comparing corresponding components, we get

$$\begin{bmatrix} 2 & 4 & 0 & b_1 \\ -3 & 1 & -7 & b_2 \\ 1 & 1 & 1 & b_3 \end{bmatrix}$$

$\underbrace{\hspace{10em}}_A$

01 mark

$$\det(A) = 0$$

01 mark

(Using theorem 2.3.8, v_1, v_2, v_3 are linearly dependent. 01 mark
 A is not invertible
 $AX=b$ is inconsistent)

$\{v_1, v_2, v_3\}$ is not a basis for $\mathbb{R}^3$. 01 mark

Q2(b):

$$x_1 + 5x_2 + x_3 + 2x_4 - x_5 = 0$$

$$x_1 - 2x_2 - x_3 + 3x_4 + 2x_5 = 0$$

$$\begin{bmatrix} 1 & 5 & 1 & 2 & -1 \\ 0 & 1 & 2/7 & -1/7 & -3/7 \end{bmatrix}$$

01 mark

1- Solution $(x_1, x_2, x_3, x_4, x_5) = \left(\frac{3}{7}r - \frac{19}{7}s - \frac{8}{7}t, -\frac{2}{7}r + \frac{1}{7}s + \frac{3}{7}t, s, s, t \right)$

$$(1, 5, 1, 2, -1) \cdot (x_1, x_2, x_3, x_4, x_5) = 0$$

$$x_3 = s, x_4 = s, x_5 = t$$

$$(1, -2, -1, 3, 2) \cdot (x_1, x_2, x_3, x_4, x_5) = 0$$

$$s, s, t \in \mathbb{R}$$

01 mark

$$(x_1, x_2, x_3, x_4, x_5) = r \underbrace{\left(\frac{3}{7}, -\frac{2}{7}, 1, 0, 0 \right)}_{\vec{w}_1} + s \underbrace{\left(-\frac{19}{7}, \frac{1}{7}, 0, 1, 0 \right)}_{\vec{w}_2} + t \underbrace{\left(-\frac{8}{7}, \frac{3}{7}, 0, 0, 1 \right)}_{\vec{w}_3}$$

0.5 mark

basis

$$\text{for Solution Space} = \{ \vec{w}_1, \vec{w}_2, \vec{w}_3 \}$$

$$\text{Dimension " " } = 03$$

0.5 mark

$$\text{Nullity} = 03$$

- Basis for Row Space = $\{ \vec{r}_1 = (1, 5, 1, 2, -1), \vec{r}_2 = (0, 1, 2/7, -1/7, -3/7) \}$

" " Column Space = $\{ \vec{c}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \vec{c}_2 = \begin{bmatrix} 5 \\ -2 \end{bmatrix} \}$

01 mark

$$\text{Rank} + \text{Nullity} = 5 \quad \text{or} \quad \text{Nullity} = \text{Dim for sol space} = 03$$

$$\text{Rank} = \text{ " " Row/col Space} = 02$$

01 mark

Q3: $B = \{(1,2), (2,3)\}$, $B' = \{(1,3), (1,4)\}$

For $P_{B \rightarrow B'}$; $\left[\begin{array}{cc|cc} 1 & 1 & 1 & 2 \\ 3 & 4 & 2 & 3 \end{array} \right] \rightarrow \underline{(02 \text{ marks})}$

Reduced Row Echelon form

$$\left[\begin{array}{cc|cc} 1 & 0 & 2 & 5 \\ 0 & 1 & -1 & -3 \end{array} \right]; P_{B \rightarrow B'} = \begin{bmatrix} 2 & 5 \\ -1 & -3 \end{bmatrix}$$

(03 marks)

For $P_{B' \rightarrow B}$ $\left[\begin{array}{cc|cc} 1 & 2 & 1 & 1 \\ 2 & 3 & 3 & 4 \end{array} \right] \rightarrow \underline{(02 \text{ marks})}$

Reduced Row Echelon form

$$\left[\begin{array}{cc|cc} 1 & 0 & 3 & 5 \\ 0 & 1 & -1 & -2 \end{array} \right] \quad \underline{(02 \text{ marks})}$$

$$P_{B' \rightarrow B} = \begin{bmatrix} 3 & 5 \\ -1 & -2 \end{bmatrix} \quad \underline{(01 \text{ marks})}$$